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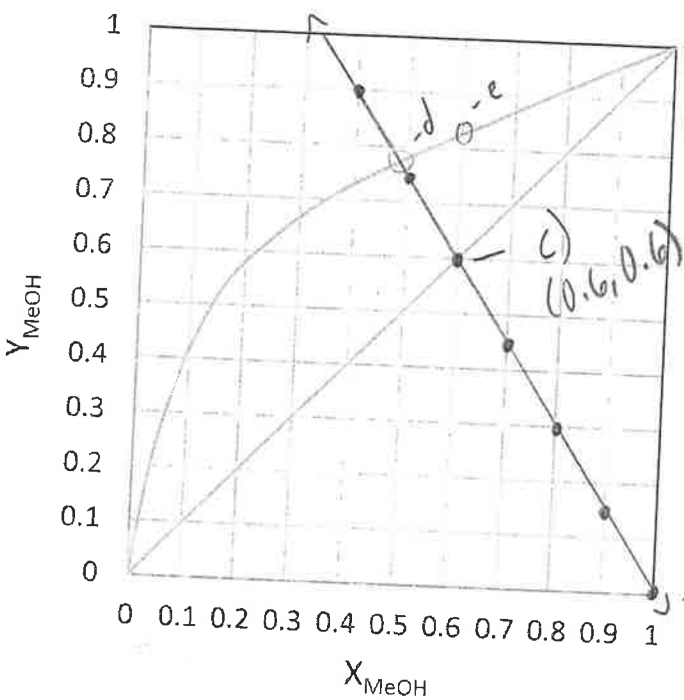
CHE 313 Transport Phenomena III

Quiz #1 (9/16/25) **Name:** Lucia Hernandez

1. We are separating a mixture of methanol and water in a flash drum at 1.0 atm pressure. Feed is 60.0 mol% methanol, and 40.0 % of the feed is vaporized. Feed flow rate is 100 kmol/hr. z

Equilibrium xy graph is:

x	y
0	3/2
0.1	1.35
0.2	1.2
0.3	1.05
0.4	0.9
0.5	0.75
0.6	0.6
0.7	0.45
0.8	0.3
0.9	0.15
1	0



$y = -3/2\, x + \frac{3}{2}$ (operate line)

a) $F = 100$ kmol/hr, $z = 0.6$ → V, y ; liquid → L, x

$F = L + V$ - vapor

$Fz = Lx + Vy$

$\frac{V}{F} = 0.4$

a) (2pt) Sketch the process, define all variables, and set up overall mass balance and component mass balance equations.

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b) (2pts) Derive an operating line equation using component mass balance equations (don't just write down the operating equation).

b) $Fz = Lx + Vy$

$Vy = Lx + Fz$

$y = -\frac{L}{V}x + \frac{F}{V}z$

$\frac{V}{F} = 0.4 \rightarrow V = 0.4F = 40$ (with $F = 100$)

* $y = -\frac{L}{V}x + \frac{F}{V}z$

$y = -\frac{3}{2}x + \frac{3}{2}$

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c) (2pts) Find liquid flow rate (L), vapor flow rate (V), a slope of the operating line, and a point on y=x curve that operating line should pass.

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* $V = (0.4)(100) = 40$ kmol/hr

* $L = F - V = 100 - 40 = 60$ kmol/hr

* slope: $\frac{L}{V} = \frac{60}{40} = \frac{3}{2}$

$\frac{F}{V} = \frac{100}{40} = \frac{5}{2}$

* $y = -\frac{3}{2}x + (5/2)(0.6) = -\frac{3}{2}x + 3/2$

d) (2pts) Draw the operating line on the graph, and find the vapor (y) and liquid (x) mole fractions.

(0.49, 0.77)
x y

\# where op line & eq curve intersect

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e) (2pts) By changing the amount of evaporated mixture, we want to achieve the maximum possible methanol content in the distillate. Use graphical solution method.

$Fz = Lx + Vy = \cancel{F}_x\, x + \cancel{F}_y\, y$ - max distillate wanted $\rightarrow z = x = 0.6$

$y_{max} = 0.82$

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